FEVEN ABEBE BEJIGA

MSc Artificial Intelligence | AI/ML Research | Agricultural AI
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Portfolio | GitHub | LinkedIn

PROFESSIONAL SUMMARY

MSc Artificial Intelligence student at Addis Ababa University with a BSc in Health Informatics from the University of Gondar (CGPA 3.86/4.00). Brings 3+ years of transferable technical and implementation experience in digital health systems, data management, and software development. Co-author of a federated learning paper on metaheuristic client selection, submitted to PanAfriCon AI 2026. Developing AgriMind, an early-stage initiative on maize and wheat disease information for agricultural extension workers in Ethiopia. Interested in reliable, resource-aware, and context-sensitive AI for agriculture and sustainable development.

EDUCATION

MSc in Artificial Intelligence — Addis Ababa University (AAiT) November 2024 – Present

BSc in Health Informatics — University of Gondar 2022
CGPA: 3.86/4.00

RESEARCH EXPERIENCE AND PUBLICATIONS

"Lightweight Metaheuristic Optimization for Client Selection in Resource-Constrained Federated Learning Systems"
Feven Abebe Bejiga, Zehara Yassin, and Beakal Gizachew (Addis Ababa University)
Submitted to PanAfriCon AI 2026

- Investigated Genetic Algorithm (GA), Particle Swarm Optimization (PSO), and Simulated Annealing (SA) as client-selection strategies in federated learning, compared against random selection using FedProx on FEMNIST, CIFAR-10, and CIFAR-100.
- Experimental design: five seeds and 100 communication rounds per configuration. Built the PyTorch framework and experimental pipeline (see Selected Projects).

SELECTED PROJECTS

AgriMind — Agricultural AI Initiative *(early-stage)*
Python · Image Classification · Agricultural Knowledge Retrieval

- **Problem:** helping agricultural extension workers access useful, reliable maize and wheat disease information in resource-constrained settings.
- **Work so far:** exploring AI-based crop-disease identification from images and retrieval of relevant agricultural information, and preparing agricultural reference material for an early retrieval proof of concept.
- **Direction:** developing reliable, context-sensitive AI for Ethiopian agriculture. Not yet field-tested or validated with agricultural experts.

Federated Learning with Metaheuristic Client Selection
Python · PyTorch · FedProx · GA / PSO / SA

- Built a federated learning framework for non-IID client data using PyTorch and FedProx, with a multi-client pipeline covering local training, global aggregation, and iterative communication rounds.
- Evaluated GA-, PSO-, and SA-based client selection against a random baseline on training loss and classification accuracy.

House Price Prediction System
Python · scikit-learn · CatBoost · FastAPI · Streamlit · Docker · GitHub Actions

- Built an end-to-end ML system for real-estate price prediction: data preprocessing, feature engineering, a FastAPI REST API for serving predictions, and a Streamlit front end.
- Containerized the application with Docker and added automated testing with GitHub Actions CI.

Fraud Detection Machine Learning Project
Python · scikit-learn · Logistic Regression · Random Forest

- Compared Logistic Regression and Random Forest classifiers on highly imbalanced fraud data, using F1-score and precision-recall AUC (AUC-PR) as imbalance-appropriate metrics.

ROS 2 Autonomous Lane-Following Robot (simulation)
ROS 2 Humble · Gazebo · Python · Computer Vision · PID control

- Developed a lane-following robot simulation with HSV-based lane detection and PID steering control, integrating perception and control in a single ROS 2 pipeline.

PROFESSIONAL EXPERIENCE

Digital Health Developer — CDHi, University of Gondar July 2023 – January 2026

- Customized EMR and DHIS2 systems to fit hospital workflows and organizational requirements, working with clinical and administrative stakeholders.
- Implemented WHO Digital Adaptation Toolkit (DAK) content within the University of Gondar Referral Hospital's EMR environment, supporting alignment with national digital health standards.
- Maintained the MySQL databases behind the hospital's digital health systems, ran data integrity checks, and produced reports from EMR and DHIS2 data for decision-making and reporting.
- Developed DHIS2 Tracker tools for Health and Demographic Survey activities and aggregate data-collection tools for ISHO, supporting structured data collection, management, and analysis.

Fellow — Center for Digital Health and Implementation (CDHi) November 2022 – June 2023

- Supported digital health and health information systems activities.
- Gained foundational experience in digital health technologies and health information management.

TECHNICAL SKILLS

Programming	Python, SQL, C++, Java, JavaScript
CS Foundations	Data structures, algorithms, object-oriented programming, software engineering fundamentals
ML and Deep Learning	PyTorch, TensorFlow, scikit-learn, supervised learning, model evaluation, neural networks, CNNs, Transformers
NLP and Retrieval	Natural Language Processing, Retrieval-Augmented Generation
Data Science	pandas, NumPy, SciPy, Matplotlib, EDA, data preprocessing, statistical analysis, time-series analysis
Computer Vision	Image processing, image classification, OpenCV
Robotics and Control	ROS 2, Gazebo, vision-based lane detection, PID control, robotics simulation
Engineering and Deployment	Git, GitHub, FastAPI, REST APIs, Docker, GitHub Actions, Streamlit
Data and Digital Systems	MySQL, database management, data integrity, EMR, DHIS2, health information systems

ADDITIONAL TECHNICAL TRAINING

Data Science and AI Engineering — Qiyas, Addis Ababa University May – September 2026 (completed)

SAFEE KAIM — Kifiya AI Mastery Training Program, 10 Academy Women-Only Cohort 2026

GCI World April 2026 — Data Science, Matsuo-Iwasawa Laboratory, Graduate School of Engineering, The University of Tokyo 2026

LANGUAGES

Amharic — Native | **English** — Fluent